

Sandeepa Chathumina Karunanayake

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GitHub: <https://github.com/SandeepaChathumina>

Portfolio: <https://my-portfolio-murex-xi-12.vercel.app/>

PROFESSIONAL SUMMARY

Software Engineering undergraduate specializing in Full-Stack and Distributed System Development with hands-on experience in MERN stack, Microservices Architecture, Docker, Kubernetes, and RESTful APIs. Skilled in building scalable web applications, authentication systems, and cloud-ready services with strong knowledge of software engineering principles. Achieved Dean's List recognition in Year 2 Semester 2 with a GPA of 3.73. Passionate about backend engineering, scalable architectures, and modern web technologies.

TECHNICAL SKILLS

- Languages: Java, JavaScript, SQL
 - Frontend: React.js, Next.js, Tailwind CSS
 - Backend: Node.js, Express.js
 - Mobile Development: Kotlin (Android)
 - Databases: MongoDB, MySQL
 - Microservices & DevOps: Docker, Kubernetes, REST APIs, JWT Authentication
 - Tools & Platforms: Git, GitHub, Postman, VS Code
 - Concepts: Microservices Architecture, Distributed Systems, RBAC, API Integration
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ACADEMIC PROJECTS

<https://github.com/SandeepaChathumina/AgriHUB-LK>

AgriHUB-LK | Agricultural Supply Chain Management System

Role: Team Leader | MERN Stack Group Project

- Led the team in developing a full-stack agricultural supply chain platform using the MERN Stack.
- Contributed to the initial project setup, including backend/frontend structure, routing, and environment configuration.
- Developed the Order Management module to support crop/product ordering, order tracking, and status updates.
- Implemented the Payment Management module to handle payment-related workflows for orders.
- Developed the Message Service to support communication between users such as farmers, distributors, transporters, and admins.
- Integrated REST APIs between frontend and backend modules for smooth data handling and user interaction.
- Coordinated task allocation, Git/GitHub collaboration, module integration, and project progress as team leader.

<https://github.com/SandeepaChathumina/smart-healthcare-platform>

Smart Healthcare Platform | Distributed Microservices System

Role: Team Leader | Group Project

- Led the team in designing and developing a distributed healthcare platform using Microservices Architecture.
- Contributed to Docker setup and containerization to support service-based deployment.
- Developed the Appointment Service for managing doctor availability, patient bookings, and appointment workflows.
- Implemented the Payment module to support appointment-related payment processing.
- Developed the Telemedicine feature to support online doctor-patient consultation functionality.
- Integrated REST APIs and supported inter-service communication between healthcare services.

- Collaborated using Git/GitHub and coordinated task distribution, module integration, and project progress as team leader.

<https://github.com/SandeepaChathumina/Bus-Rental-Management-System-Bus-Zone->

BusZone+ | Bus Rental Management System

Role: Team Leader | MERN Stack Group Project (5 Members)

- Led the development of a full-stack bus rental management platform using the MERN Stack.
- Developed User and Booking Management modules with JWT-based authentication and role-based access control.
- Integrated Google Places API to build an AI-powered travel recommendation feature for routes and accommodations.
- Implemented an AI chatbot using Ollama to automate Frequently Asked Questions (FAQs).
- Coordinated Git/GitHub collaboration, task management, and module integration within the development team.

<https://github.com/SandeepaChathumina/Health-Routine-App>

Health Flow | Health & Wellness Tracker (Mobile)

- Developed a native Android wellness application using Kotlin with responsive UI and custom animations.
- Implemented features including habit tracking, hydration reminders, BMI calculation, and mood monitoring.
- Integrated phone sensor functionality and local data persistence using Shared Preferences.

ACHIEVEMENTS

- Dean's List Nominee – Year 2 Semester 2 (GPA: 3.73)

EDUCATION

BSc (Hons) in Information Technology (Specializing in Software Engineering)

Sri Lanka Institute of Information Technology (SLIIT)

Expected Graduation: 2027

- **Current GPA: 3.59**
- **Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming , Web Application Development, Software Engineering, Distributed Systems.**

CERTIFICATIONS

- **AI/ML Stage 1 & 2 – SLIIT (Sri Lanka Institute of Information Technology)**
- **Web Development Certification – University of Moratuwa**
- **Front-End Development Certification – University of Moratuwa**